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DS 120 Pop Quiz 4

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Instructions

- Unless otherwise instructed by a Teaching Fellow, the time limit for this quiz is **10 minutes**. Please manage your time accordingly.
- At the end of the quiz, please **submit** your completed paper to your Teaching Fellow.
- Only tools required for writing your answers are allowed on the desk. All other items must be put away.

Section 1 - Multiple Choice

Select only one option

Question 1. If

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}, \quad \left| \begin{array}{cc|cc} & & 1 & 0 \\ & & 0 & 1 \end{array} \right.$$

then the inverse of A is:

- (A) $\frac{1}{-1} \begin{bmatrix} 2 & -3 \\ -3 & 4 \end{bmatrix}$
- (B) $\frac{1}{-1} \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$
- (C) $\frac{1}{-1} \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$
- (D) $\frac{1}{1} \begin{bmatrix} 2 & -3 \\ -3 & 4 \end{bmatrix}$

$$\left| \begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 3 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & -2 & -1 \\ 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & -1 \end{array} \right.$$

Question 2. The reduced row echelon form (RREF) of a matrix is:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}.$$

Which of the following statements is true?

- (A) The matrix has rank 1 and two pivot columns.
- (B) The matrix has rank 2 and two linearly independent columns.
- (C) The matrix has rank 3 and three linearly independent columns.
- (D) The matrix has rank 2 and three linearly independent columns.

Section 2

Given

$$f(x, y) = x^2 + y^2 + xy$$

Is $(0, 0)$ a saddle point?

(Note: It is sufficient to show that $(0, 0)$ is a critical point, and determine if the statement is true or false with appropriate reasoning.)

$$\begin{aligned} f_x &= 2x + y & f_x(0) &= 0 \\ f_y &= 2y + x & f_y(0) &= 0 \end{aligned} \rightarrow \text{critical point}$$

$$\left. \begin{aligned} f_{xx} &= 2 \\ f_{yy} &= 2 \\ f_{xy} &= 1 \\ f_{yx} &= 1 \end{aligned} \right\} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow 2 \cdot 2 - 1 = 3 > 0 \rightarrow \text{saddle point}$$